

Final Project Execution Report by Data Science

Project: AI-Based Personalized Diet & Recovery Planner

1. Objective

Prepare clean, validated, and ML-ready healthcare datasets for the AI/ML team by performing preprocessing, feature engineering, relationship validation, and dataset preparation.

2. Workflow Executed

Raw Datasets → Preprocessing → Missing Value Validation → Encoding & Scaling Assessment → Data Quality Validation → Feature Engineering → Patient-Tracking Relationship Verification → ML-Ready Dataset Generation

3. Tasks Completed

1. Implement Preprocessing Pipeline

- `load_data()`
- `handle_missing_values()`
- `remove_duplicates()`
- `validate_data()`
- `verify_relationships()`
- `feature_engineering()`
- `save_processed_data()`

2. Handle Missing Values

Performed missing value analysis and verified dataset completeness.

3. Encoding and Scaling

Evaluated categorical encoding and feature scaling requirements for ML models.

4. Validate Data Quality

Checked data consistency, duplicates, ranges, and overall integrity.

5. Create Engineered Features

Generated healthcare-focused features such as BMI Category, Age Group, Average Blood Sugar, Average Daily Steps, and Average Sleep Hours.

6. Verify Patient-Tracking Relationships

Verified referential integrity between patient and tracking datasets.

7. Generate Final ML-Ready Datasets

Prepared final processed datasets for AI/ML model development.

4. Feature Engineering Implementation

8. BMI_Category
9. Age_Group
10. Avg_Blood_Sugar
11. Avg_Daily_Steps
12. Avg_Sleep_Hours

5. Final Feature List

Feature	Type
Age	Numerical
Gender	Categorical
Height_cm	Numerical
Weight_kg	Numerical
BMI	Numerical
BMI_Category	Engineered
Age_Group	Engineered
Avg_Blood_Sugar	Engineered
Avg_Daily_Steps	Engineered
Avg_Sleep_Hours	Engineered

6. Deliverables

13. processed_patients.csv
14. processed_tracking.csv
15. Feature Engineering Implementation
16. Final Feature List

7. Handover to AI/ML Team

The processed datasets and engineered features are ready for diet recommendation engine development, recovery analysis, recommendation generation, and alert generation modules.